

Anup Rai

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SUMMARY

Final-year Electronics & Communication Engineering student specializing in applied machine learning and data analytics. Experienced in building real-time intelligent systems using probabilistic modeling, gesture-based interaction, and LLM-powered applications. IEEE Best Paper Award recipient (AI & ML Track, 2026).

TECHNICAL SKILLS

Programming Languages: Python, C, C++

Machine Learning: Supervised/Unsupervised Learning, K-Means, Scikit-learn, NumPy, Pandas

Probabilistic Models: Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT)

Computer Vision: OpenCV, MediaPipe

Tools & Platforms: Git, Jupyter, Google Colab, Unity, Ollama

Other: Graphic Design, Video Editing

PROJECTS

Inclusive Mixed Reality Math Experience Using ML-Based Models

2025 – 2026

Final Year Project

Python, OpenCV, MediaPipe, Scikit-learn

- Built a real-time gesture-controlled tutoring system integrating computer vision and adaptive ML
- Implemented Bayesian Knowledge Tracing (BKT) for adaptive mastery modeling
- Applied clustering techniques to enable personalized learning pathways

LocalPDF-QA – Local PDF Question Answering System

2026

LLM Application

Python, LLMs, Ollama, Retrieval QA

- Built a fully local PDF-based question-answering system using open-source LLMs via Ollama
- Implemented document parsing and retrieval pipeline for efficient knowledge extraction

MediaPipe Cube Control via Hand Gesture

2026

Computer Vision / HCI

Unity, C#, MediaPipe

- Developed a low-latency real-time hand-tracking interface using MediaPipe integrated with Unity
- Designed modular gesture-state architecture for scalable gesture recognition

RESEARCH

DKT-Based AR Framework for Predictive Learning Assessment

2025 – 2026

IEEE ICTP 2026 | Best Paper Award – AI & ML Track

- Designed an adaptive Augmented Reality system using Deep Knowledge Tracing (DKT)
- Developed system architecture following Research-Through-Design (RTD) methodology
- Achieved Accuracy: 0.84 and PR-AUC: 0.82 in real-world classroom deployment

EDUCATION

Nepal Engineering College, Pokhara University

Bhaktapur, Nepal

B.E. in Electronics & Communication Engineering

2021 – 2026

Pokhara Aadarsha Secondary School

Pokhara, Nepal

Science (Grade 11 – 12)

2019 – 2021

CERTIFICATIONS & HONORS

- **Best Paper Award** — IEEE ICTP 2026 (AI & ML Track)
- Data Science with Python — Code IT
- IEEE Student Volunteer